



CAN
(Controller Area Network)
Generic Diagnostic Specification

Worldwide Requirements

Version 2003.0
Date Issued: 25 April 2003

Research & Vehicle Technology Office
Electrical/Electronic Systems Engineering
Core Network Communications

FORD MOTOR COMPANY CONFIDENTIAL AND PROPRIETARY

This document contains Ford Motor Company confidential and proprietary information. Disclosure of the information contained in any portion of this document is not permitted without the expressed, written consent of a duly authorized representative of Ford Motor Company, Dearborn, Michigan, USA.

© Copyright 1998 – 2003, Ford Motor Company - All Rights Reserved

REVISION HISTORY

Previous Version	Current Version	Version Description	Author	Date
N/A	v2000.0	Initial Release	S. DiLodovico	21- July - 2000
v2000.0	v2000.1	Clarification and minor alterations.	S. DiLodovico	27-July - 2000
v2000.1	v2000.2	Changed and clarified Response code \$78 handling	S. DiLodovico	28-July - 2000
v2000.2	v2001.0	Clarification and additions to diagnostic services and references.	S. DiLodovico / J. Miller	09-February-2001
v2001.0	v2001.1	Inclusion of Errata sheet details into the specification and clarification of issues.	J. Miller	30-November-2001
v2001.1	v2003.0	Inclusion of Errata sheet details into the specification and clarification of issues.	J. Miller	25-April-2003

Release	Previous Section	Current Section	Change Description
			where the parameters are defined.
v2003.0	Table 45	Table 46	Changed N_Bs timeout value from 75ms to a range of 75ms – 120ms
v2003.0	Table 45	Table 47	Changed N_Cr timeout value from 75ms to a range of 75ms – 120ms
v2003.0	A.2	A.2	Clarification on usage of NRCs \$21 and \$78
v2003.0	A.2.17	A.2.17	Added an additional reason of incorrect dataFormatIdentifier to negative response code \$31 for service \$34 (requestDownload)
v2003.0	A.2.18	A.2.18	Added an additional reason of incorrect dataFormatIdentifier to negative response code \$31 for service \$35 (requestUpload)
v2003.0	A.2.21	A.2.21	Added an additional negative response code of \$10 (generalReject) to service \$3B (writeDataByLocalID).
v2003.0	A.2.25	A.2.25	Added an additional reason of requested data rate does not match active data rate to negative response code \$22 (conditionsNotCorrect) for service \$A0 (requestDiagnosticDataPacket).
v2003.0	A.2.27	A.2.27	Added an additional reason of all prerequisite conditions are not satisfied to negative response code \$22 (conditionsNotCorrect) for service \$B1 (diagnosticCommand).

2.2.2.7.23.3	General Reporting Requirements	35
2.2.2.7.23.4	Periodic Positive Responses	36
2.2.2.8	Diagnostic Gateway Requirements	36
2.2.2.8.1	Purpose / Scope	36
2.2.2.8.2	Entering Gateway State	37
2.2.2.8.3	Gateway State Message Handling	37
2.2.2.8.3.1	Messages: Tester to Gateway	37
2.2.2.8.3.2	Messages: Gateway to Satellite	38
2.2.2.8.3.3	Messages: Satellite to Gateway	38
2.2.2.8.4	Gateway State Timing Requirements	38
2.2.2.8.5	Satellite Modules	38
2.2.2.8.6	Satellite Module in a Diagnostic Session	38
2.2.2.8.7	Satellite Module Mandatory PIDs	38
2.2.2.8.8	Redirecting Communication	38
2.2.2.9	CAN Diagnostic Specification Version CommonID/PID (ref. P1-v2003.0, 16.7)	39
2.2.3	Additional Requirements	40
2.2.3.1	Diagnostic Services	40
2.2.3.1.1	communicationControl (28 hex) service	40
2.2.3.1.1.1	Request Message Definition	40
2.2.3.1.1.1.1	Request Message Sub-Function Parameter LEV_ Definition	40
2.2.3.1.1.1.2	Request Message Data Parameter Definitions	40
2.2.3.1.1.2	Positive Response Message Definition	41
2.2.3.1.2	controlDTCSetting (85 hex) service	41
2.2.3.1.2.1	Service Description	41
2.2.3.1.2.2	Request Message Definition	41
2.2.3.1.2.2.1	Request Message Sub-Function Parameter \$Level (LEV_) Definition	41
2.2.3.1.2.3	Positive Response Message Definition	41
2.2.3.1.2.3.1	Positive Response Message Data Parameter Definition	41
2.2.3.1.3	requestCommonIDScalingMasking (24 hex) service	42
2.2.3.1.3.1	Service Description	42
2.2.3.1.3.2	Request Message Definition	42
2.2.3.1.3.3	Positive Response Message Definition	42
2.2.3.1.3.3.1	Positive Response Message Data Parameter Definition	42
2.2.3.2	Network Fault Strategy	43
2.2.3.2.1	Missing Message Network Faults	43
2.2.3.2.2	Invalid Data Faults	43
2.2.3.2.3	Bus Off Faults	44
2.2.3.2.4	Private Network Faults	44
2.2.3.3	Reporting of Logged DTCs	44
2.2.3.4	Negative Response Messages (Service \$7F)	44
2.2.3.5	Safety and Damage Prevention Requirements	44
2.2.3.6	Powertrain KOEO Selftest \$02	45
2.2.3.7	Intermessage Gap Requirements	45
2.2.3.8	Timed Operation Timeout	45
2.2.3.9	Memory Modification and Output Control	46
2.2.3.10	Local ID Differentiation	47
2.2.3.11	Diagnostic Mode (PID/common ID)	47
2.2.3.12	Module Configuration and Programming	47
2.2.3.13	Special Procedures and Requirements	47
2.2.3.14	Powertrain Specific Requirements	47
2.2.3.15	FNOS Mandatory PIDs	48
2.2.3.16	Inter-module Communication Gateway	48
2.3	NETWORK LAYER	49
2.3.1	Referenced Requirements	49
2.3.1.1	Addressing Format (ref ISO 15765-2, 5.2, 7.3.5.1, 7.4.1, 7.4.2, 8.3)	49

LIST OF TABLES

Table 1: Applicable Documents Table	10
Table 2: Diagnostic Application Layer Informational References	15
Table 3: Diagnostic Application Layer Referenced Requirements	15
Table 4: Service Identifier Value Summary Table	17
Table 5: startDiagnosticSession diagnosticMode Values Supported	19
Table 6: readMemoryByAddress.....	21
Table 7: writeMemoryByAddress Request Message.....	21
Table 8: writeMemoryByAddress Positive Response Message	21
Table 9: readDiagnosticTroubleCodesByStatus statusOfDTC Values Supported	24
Table 10: Allowable inputOutputControlParameter values.....	26
Table 11: Allowable RETO_ values.....	27
Table 12: Allowable RETS_ values	27
Table 13: Allowable routineStatus Byte values.....	28
Table 14: Allowable Routine Status Byte values	28
Table 15: requestDownload Service Message.....	28
Table 16: requestDownload Positive Response Service Message.....	29
Table 17: requestUpload Service Message.....	29
Table 18: requestUpload Positive Response Service Message.....	29
Table 19: testerPresent Request Message.....	32
Table 20: diagnosticCommand Request Message	32
Table 21: diagnosticCommand Positive Response Message	32
Table 22: diagnosticCommand Negative Response Message	33
Table 23: dynamicallyDefineDiagnosticDataPacket Message	33
Table 24: DPID Format Byte (DPIDFB) Parameter Values.....	33
Table 25: dynamicallyDefineDiagnosticDataPacket Positive Response Message	34
Table 26: requestDiagnosticDataPacket(s) Message.....	34
Table 27: Data Rate (DR) Parameter Values.....	35
Table 28: requestDiagnosticDataPacket(s) Positive Response Message.....	35
Table 29: CAN Diagnostic Specification Version CommonID/PID \$E6F3	39
Table 30: communicationControl Request Service Message	40
Table 31: controlByte (CB) Parameter Values	40
Table 32: communicationType (CTP) Parameter Values.....	40
Table 33: communicationControl Positive Response Message	41
Table 34: controlDTCSetting Request Message Definition	41
Table 35: controlDTCSetting Sub-function Values	41
Table 36: controlDTCSetting Positive Response Message	41
Table 37: Response Message Data Parameter Definition.....	41
Table 38: requestCommonIDScalingMasking Request Message Definition	42
Table 39: requestCommonIDScalingMasking Positive Response Message.....	42
Table 40: Response Message Data Parameter Definition.....	42
Table 41: Negative Response Data Byte Layout.....	44
Table 42: Services that Support the Lack of Diagnostic Dialog Timer.....	46
Table 43: Mandatory (PID/common ID) \$D100 States.....	47
Table 44: FNOS CommonIDs/PIDs	48
Table 45: Diagnostic Network Layer Referenced Requirements	49
Table 46: Segmented Message Performance Parameter Definitions	50
Table 47: Tester to ECU Segmented Error Handling.....	50
Table 48: ECU to Tester Segmented Error Handling.....	50
Table 49: Diagnostic Request / Response Message IDs.....	51
Table 50: Diagnostic Data Link Layer Referenced Requirements	51

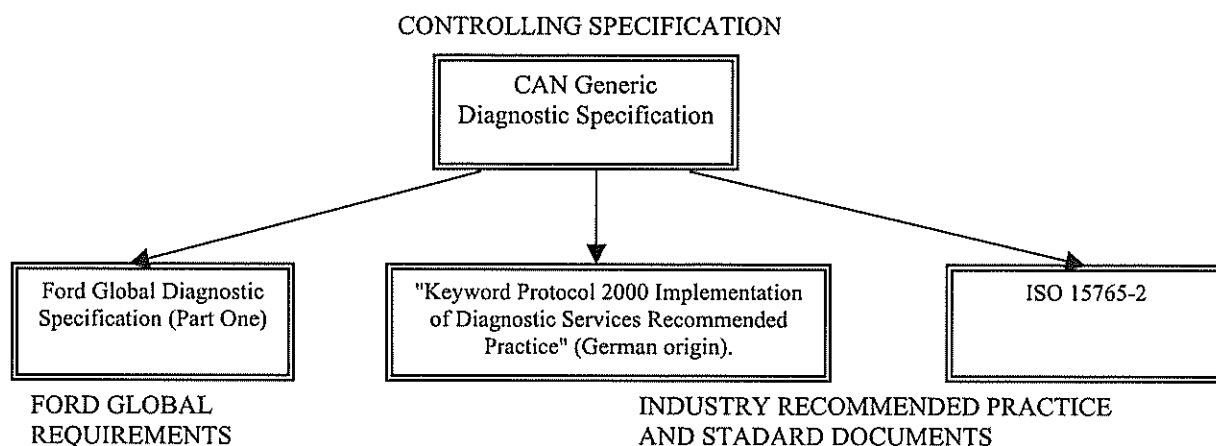


Figure 1: CAN Diagnostic Specification Relationships

This specification addresses diagnostic strategies relevant to the CAN diagnostic interface. It does not address normal or fault modes relating to the ECU to ECU communications over CAN. The "HS/MS-CAN ECU and Subsystem Requirements Specification" and "HS/MS-CAN System Level and Distributed Applications Requirements Specification" address implementation of CAN networking for normal and fault mode communications. FCSD / DSP additional requirements are contained within the Communication "Linked Based" Service Diagnostic Requirements specification (see Table 1). Additional powertrain requirements are contained within the CAN Generic Diagnostic Specification Assembly Plant Powertrain Configuration and Testing Supplement (see Table 1).

Though an attempt was made to cover all aspects of implementing CAN networks for Ford there may still be design attributes that may lack detail or are overlooked. As these and other attributes are further studied requirements will be added or modified to reflect the latest industry and corporate directions and trends.

1.4 Applicable Documents

The following documents are either referenced by this specification, or contain information that is relevant to this specification.

Table 1: Applicable Documents Table

Source	Title	Document Reference No.
Ford	CAN Generic Diagnostic Test Procedure Specification	CDTP-v2003.0
Ford	Global Diagnostic Specification (Part 1) v2003.0	P1-v2003.0
Ford	Module Programming and Configuration Specification v2003.0	MC-v2003.0
Ford	HS/MS-CAN ECU and Subsystem Requirements Specification	
Ford	HS/MS-CAN System Level and Distributed Applications Requirements Specification	
SAE	Glossary of Vehicle Networks for Multiplex and Data Communication	J1213
SAE	Electrical/Electronic Systems Diagnostic Terms, Definitions, Abbreviations, and Acronyms	J1930
SAE	Diagnostic Connector	J1962
SAE	Diagnostic Trouble Code Definitions	J2012
SAE	Enhanced E/E Diagnostic Test Modes	J2190
SAE	E/E Data Link Security	J2186
SAE	High Speed CAN (HSC) Implementation For Passenger Vehicle Applications	J2284
ISO	Road Vehicles – Diagnostics on CAN – Part 2: Network layer	15765-2

1.5 Abbreviations / Acronyms

The following abbreviations are used throughout this specification:

CAN:	Controller Area Network
DLC:	Data Link Connector (SAE J1962)
DSP:	Diagnostic Service Planning
ECU:	Electronic Control Unit
E/E:	Electrical and Electronic
EESE:	Electrical/Electronic Systems Engineering
FAO:	Ford Automotive Operations
FCSD:	Ford Customer Service Division
FNOS:	Ford Network Operating System
GDS:	Global Diagnostic Specification
ID:	Identifier
ISO:	International Standards Organization
MRDB:	Master Reference Data Base
NGS:	New Generation Star Tester
OSI:	Open System Interconnect
PCM:	Powertrain Control Module
SAE:	Society of Automotive Engineers
SSDS:	Subsystem Specific Diagnostic Specification (Part 2)
ECATS:	End-Of-Line Configuration and Test System
WDS:	Worldwide Diagnostic System
R&VT:	Research and Vehicle Technology
WCR:	Worldwide Customer Requirements
OBD:	Onboard Diagnostics
EOBD:	European Onboard Diagnostics
eCATS:	Electronic Consumer Acceptance Test System
CM:	Continuous Monitor
M:	Mandatory
O:	User Optional; support is application specific or dependent on dynamic usage
VO:	Vehicle Operations

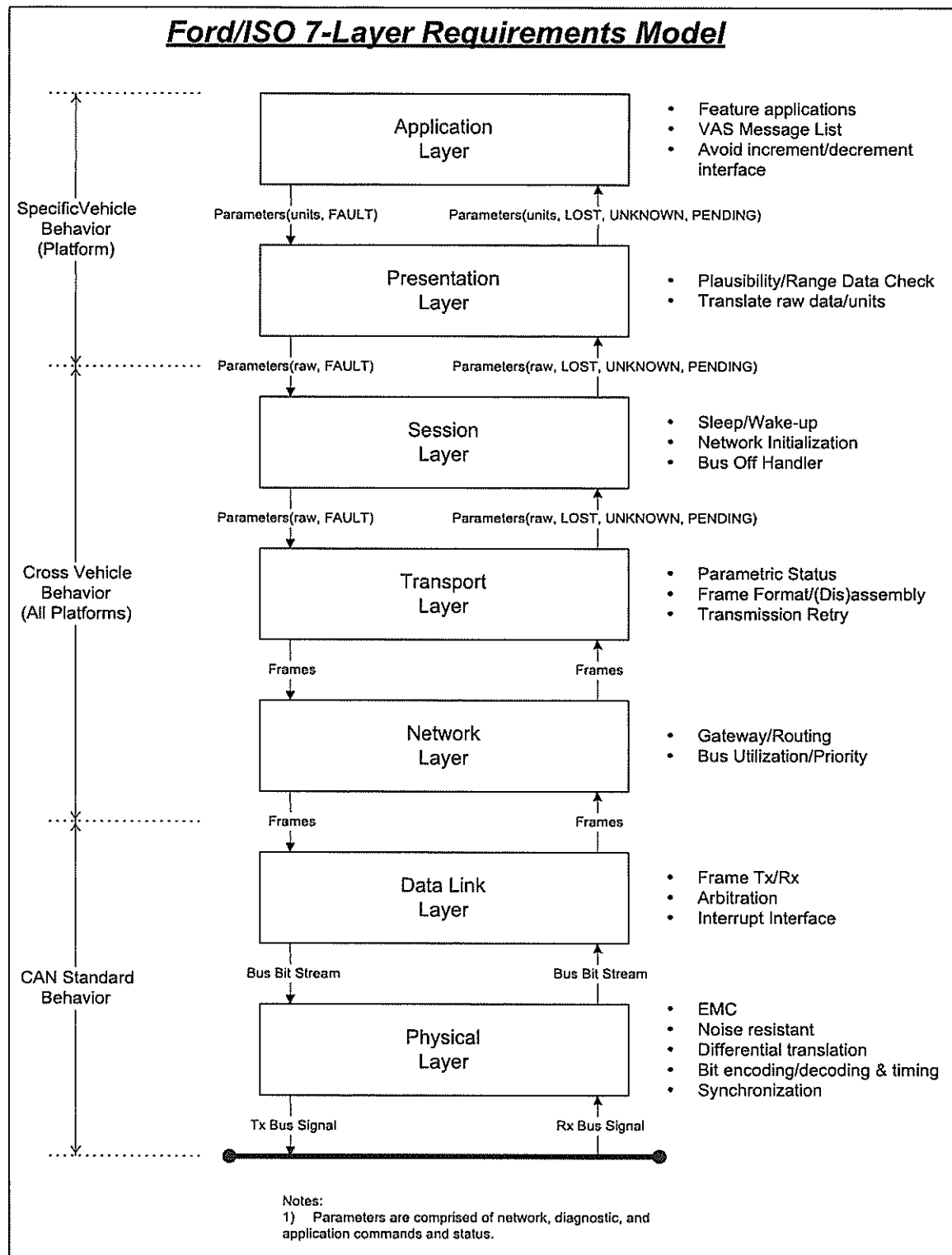


Figure 2: Ford/ISO 7-Layer Requirements Model

[27]	KWP-GRP-1.5	7.7. writeMemoryByAddress service	All except 7.7.5.1 and 7.7.5.2
[28]	KWP-GRP-1.5	8.2. readDiagnosticTroubleCodesByStatus service	All except 8.2.1.1.2, Fig. 4, Fig. 5, 8.2.5.2.3
[29]	KWP-GRP-1.5	8.4. readFreezeFrameData service	All except 8.4.5.1
[30]	KWP-GRP-1.5	8.5. clearDiagnosticInformation service	All except 8.5.5
[31]	KWP-GRP-1.5	9.2. inputOutputControlByCommonIdentifier service	All except 9.2.5
[32]	KWP-GRP-1.5	10. Remote Activation of Routine functional unit	All except 10.2, 10.3.5, 10.4, 10.5.5.2 and 10.6
[33]	KWP-GRP-1.5	11.1. requestDownload service	All except 11.1.5
[34]	KWP-GRP-1.5	11.2. requestUpload service	All except 11.2.5
[35]	KWP-GRP-1.5	11.3. transferData service	All except 11.3.4 and 11.3.5
[36]	KWP-GRP-1.5	11.4. requestTransferExit service	All except 11.4.1 and 11.4.5

2.2.2.1 Automatic Clearing of DTCs (Alternative method) (ref. P1-v2003.0, 6.3.2.1)

Continuous DTCs may be optionally cleared using a time based aging counter. The time counter shall start at zero and begin counting upward as soon as the fault that has been validated is not detected by the ECU anymore. The counter shall only be incremented while the module is in operation, as defined in its SSDS. The DTC shall be aged (deleted) after 75 hours of operation has occurred since the last validation of the logged DTC.

If the fault is detected while the counter is being incremented then the counter shall be reset to zero.

2.2.2.2 Fault PID and Fault PID DTC Requirements (ref. P1-v2003.0, 16.9, 10.4.4)

Every fault DTC defined in the MRDB shall have the following prefix added to the description: "Fault DTC:".

Every fault PID defined in the MRDB shall have the following prefix added to the description: "Fault PID (CM):" or "Fault PID (OD):".

2.2.2.3 Inapplicable K-Line Protocol References (ref. KWP-GRP-1.5, All Sections)

No K-Line references in the KWP-GRP-1.5 shall be applicable to this specification. All the network and data link layer definitions shall be referenced in Section 2.4 of this document that reference the ISO 15765-2 document referring to CAN protocol elements. Also, all references to the ISO 14230-2 specification are not applicable to any Ford requirements.

2.2.2.4 Header Bytes (ref. KWP-GRP-1.5, All Sections)

Header formats specified in KWP-GRP-1.5 are not applicable to the Ford CAN diagnostic messaging definitions. See Section 2.3.2 for information relating to CAN identifiers and references to the messaging formats for CAN. Included in this omission are all references to format, target, source and length bytes.

2.2.2.5 Request Messages/Services

The modes of operation and the services supported in each of those modes are listed in Table 4.

2.2.2.6 Response Code \$78 Logic (ref. KWP-GRP-1.5, 5.3.1.4.1)

Refer to Annex A.2 for conditions where response code \$78 may be used.

When utilizing response code \$78 logic, the initial negative response with response code \$78 shall be sent according to the timing requirements as specified in section 2.2.3.7. The timing parameters between each successive response code \$78 negative response shall not exceed 4000 ms. An ECU shall respond to a tester request with a valid positive response or negative response (other than response code \$78) within 60 seconds of the time of the original tester request. Any deviations to the 60 seconds requirement shall be documented in the ECU's Subsystem Specific Diagnostic Specification.

If the tester does not receive an additional negative response code \$78 (or the final positive or negative response), within 5 seconds of the previous negative response code \$78, or if the tester does not receive a final positive or negative response (other than response code \$78) within 60 seconds (or the time specified in the ECU's Part 2 SSDS) of the original tester request, then the tester shall assume that a response will not be received from the ECU. The tester may initiate any appropriate recovery action and begin other diagnostic sessions and strategies after the timeout period.

While the tester awaits the final positive or negative response from the ECU, it shall continually send any necessary diagnostic messages to the ECU in order to keep the lack of diagnostic dialog timer (see section 2.2.3.8) from timing out. When any valid diagnostic frame is received by an ECU during its response code \$78 handling, the ECU shall reset the appropriate timers and respond to the diagnostic frame with another service \$7F, response code \$78 message containing the same Request Service Identifier (see section 2.2.3.4) that is already being handled with the current response code \$78. Note that this Request Service Identifier is not necessarily the same as contained in the diagnostic frame causing the reset of the timers. Any deviations to this requirement shall be detailed in the ECU's Subsystem Specific Diagnostic Specification.

Note: For certain ECUs, the transmission and reception of messages is not possible while performing specific actions. For example, an ECU that is commanded to perform a flash memory erase may be unable to transmit and/or receive messages during this time. Under these circumstances, the ECU shall respond to the initial tester request with a single negative response code \$78, and respond when complete with the appropriate positive or negative response. If a second response code \$78 (or the final positive or negative response) is not received by the tester within 5 seconds, the tester shall assume that the ECU is in a no transmit/receive state. The tester shall wait a duration of 60 seconds (or the time specified in the ECU's Part 2 SSDS) for the final response. The tester shall not abort the transaction due to a lack of additional response code \$78s while waiting. This use of the single response code \$78 by an ECU for tasks that take longer than 4 seconds is only allowed while performing Method 3 programming (ref. MC-v2003.0) and requires approval by the R&VT EESE Core Network Communication Section. The specific circumstances where the single response code \$78 strategy is utilized shall be detailed in the ECU's Subsystem Specific Diagnostic Specification.

Diagnostic session \$85 (ECUProgrammingMode), if implemented, shall be used only for Method 3 file download and shall be the implementation of the Method 3 programming requirements as described in MC-v2003.0.

When an ECU transitions from any diagnostic session to another diagnostic session, the ECU shall reset all active diagnostic functionality that is not supported in the new diagnostic session (e.g., security access, I/O control), with the exception of changes written to long term memory. For example, if an ECU only supports service \$2F (inputOutputControlByCommonID) in diagnostic session \$87 then any actively controlled inputs or outputs shall revert back to the normal value as determined by the control system upon a transition from session \$87 to session \$81.

An ECU is allowed a 750ms re-initialization period upon returning to the default diagnostic session (\$81) from any other diagnostic session. During this re-initialization period the ECU is not required to respond to any diagnostic requests.

The only defined vehicleManufacturerSpecific diagnostic session is \$F0 and is defined as EOLExtendedDiagnosticSession. Any ECU that needs to grant special privileges to an End of Line test tool shall implement diagnostic session \$F0 to provide access to these privileges. For example, an ABS module that normally exits to the defaultDiagnosticSession when vehicle speed is greater than a given value may maintain the EOLExtendedDiagnosticSession (despite the vehicle speed). Diagnostic session \$F0 shall be reserved only for use by End of Line test tools in assembly plants and shall be implemented only when needed to verify communication with an End of Line tester. This EOLExtendedDiagnosticSession shall grant access to all functionality (e.g., diagnostic services, privileges, input/output control, etc.) that is necessary during the assembly testing and shall contain a superset of the diagnostic functionality supported in diagnostic session \$87 (ECUAdjustmentMode).

SystemSupplierSpecific diagnostic sessions shall not be supported by Ford test tools. All implemented systemSupplierSpecific diagnostic sessions shall be specified by the module designer and documented in the ECU's Subsystem Specific Diagnostic Specification (SSDS/Part 2).

2.2.2.7.2 ECUReset (ref. KWP-GRP-1.5, 6.5)

Only the powerOn value of the resetMode (RM_) parameter shall be supported. The resetStatus (RS_) parameter shall not be supported.

The positive response to an ECUReset (service \$11) request shall occur before the ECU performs the reset. An ECU is allowed a 750ms re-initialization period after providing a positive response to an ECUReset request. During this re-initialization period the ECU is not required to respond to any diagnostic requests.

2.2.2.7.3 clearDiagnosticInformation (ref. KWP-GRP-1.5, 8.5)

This service shall clear all groups of DTCs in the ECU, including powertrain, chassis, body and network.

Only the value of \$FF00 shall be supported for the groupOfDiagnosticInformation (GODI_) parameter, specifying that all groups shall be cleared in every clearDiagnosticInformation service request.

2.2.2.7.4 readMemoryByAddress (ref. KWP-GRP-1.5, 7.3)

This service is mainly intended for development purposes only. Most service and end-of-line equipment do not support the use of this service.

Every instance of this message shall include all four address bytes as specified in Table 6. If the ECU supports less than four address bytes, then every byte that is not used shall be set to zero in the request message with data byte #5 always corresponding to the LSB of the ECU's address.

perform secured functions is as follows:

- 1) startDiagnosticSession service
- 2) securityAccess service

The above sequence does not necessarily prohibit a tester from unsecuring an ECU before entering a given diagnostic session, but it does require the ECU to allow the tester the capability to move from any one diagnostic session to another without performing security access.

The securityAccessStatus parameter shall not be used, see Figure 3 for more details.

2.2.2.7.6.1 Seed and Key Procedure

The security access procedure shown in Figure 3 shall be used to "unlock" the ECU and grant security access. Upon receipt of a valid seed request, the ECU shall respond with a randomly chosen security seed. This seed shall remain valid or "active" in the ECU until one of the following conditions occur:

- A new validly formatted seed request is received by the ECU (regardless of whether it is for the same or a different security level and regardless of whether the security level is supported by the ECU or not).
- A security key (correct or incorrect) is received by the ECU (regardless of whether it is for the same or a different security level and regardless of whether the security level is supported by the ECU or not).
- Lack of Diagnostic Dialog timer expires, as defined in Section 2.2.3.8 of this document.

An ECU shall reject any security key sent by a test tool for a given security level if a valid or "active" seed does not exist for that security level, with a service \$7F and response code \$22.

2.2.2.7.7 readDataByCommonIdentifier (ref. KWP-GRP-1.5, 7.2)

This service is equivalent to the reading of PIDs as defined in P1-v2003.0. CommonIDs (PIDs) are not limited to a maximum of four bytes on the CAN protocol. Definitions of all CommonIdentifiers, except for those in the supplier reserved range, must be approved by the R&VT EESE Core Network Communication Section. All approved CommonIDs (PIDS) and their descriptions are defined and maintained in the MRDB.

2.2.2.7.8 readDiagnosticTroubleCodesByStatus (ref. KWP-GRP-1.5, 8.2.1.1)

The parameter values supported for the statusOfDTC (for the request message service \$18) parameter are listed in Table 9, no other parameter values shall be supported.

Table 9: readDiagnosticTroubleCodesByStatus statusOfDTC Values Supported

Hex	Description	Man / Opt	Mnemonic
00	requestIdentifiedBCDDTCAndStatus (SAE J2012 format) This value is used by the client (tester) to indicate to the server(s) (ECU(s)) that the server(s) (ECU(s)) shall respond with all diagnostic trouble codes (DTCs) which have been identified to cause a problem at the time of the request (i.e., stored in non-volatile memory).	M	RIDTCAS
01	requestSupportedBCDDTCAndStatus (SAE J2012 format) This value is used by the client (tester) to indicate to the server(s) (ECU(s)) that the server(s) (ECU(s)) shall respond with all supported diagnostic trouble codes (DTCs) regardless of their status. Even if no DTC is stored in non-volatile memory (no DTC causes a problem at the time of the request) the server shall report all DTCs which are supported. This allows the client (tester) to test the DTCReadinessFlag in the statusOfDTC parameter of each supported DTC.	O	RSUDTCAS
F0-F9	VehicleManufacturerSpecific This range of values is reserved for vehicle manufacturer specific use.	O	VMS
FA-FE	SystemSupplierSpecific This range of values is reserved for system supplier use and shall not be supported by Ford test tools.	O	SSS

A value of \$00 in the statusOfDTC in the **request** message shall trigger all codes to be reported in the positive response message that have the DTCStorageState values of either b01 (b = binary) "DTCNotPresent at time of request" or b11 "DTCPresent at time of request". The storage state of "DTCNotPresent at time of request" means that the DTC is currently being aged. The storage state of "DTCPresent at time of request" means that the fault(s) that set the DTC is still present.

Only the value of \$FF00 shall be supported for the groupOfDTC (GODTC_) parameter specifying that all groups shall be requested in every requestDiagnosticTroubleCodes service request.

The only allowable DTCFaultSymptoms for the statusOfDTC parameter in the response message \$58 are 0b0000 and 0b1111. No other values shall be allowed. Both values represent that the DTC identified does not require any additional fault symptoms (due to the specifics of the DTC descriptions in the MRDB). A value of 0b0000 shall represent that all values in the statusOfDTC parameter are supported. A value of 0b1111 shall represent that the entire statusOfDTC parameter is not supported.

NOTE: This service is only associated with requesting continuous DTCs. DTCs are not required to be reported in the same sequence as they have been detected.

Definitions of all DTCs must be approved by the R&VT EESE Core Network Communication Section. All approved DTCs and their descriptions are defined and maintained in the MRDB. A complete description of the logic used by the ECU to set each supported DTC shall be documented within the ECU's SSDS.

NOTE: If an ECU is reporting more than 255 DTCs in the response message \$58, the numberOfDTC

Case 5 Request Message: RAMI_ = \$00, FFNR_ = \$01, RI_ = Not Present in Request

- If neither DTC \$1234 nor DTC \$5678 are stored in non-volatile memory, the ECU shall return a single positive response with FFNR_ = \$01, RAMI_ = \$00, FFD = Not Present in Response, RI_ = \$0000.
- If either DTC \$1234 or DTC \$5678 are stored in non-volatile memory, the ECU shall respond with a separate positive response message for each DTC stored in non-volatile memory that supports freeze frame number \$01.

Case 6 Request Message: RAMI_ = \$00, FFNR_ = \$03, RI_ = Not Present in Request

- The ECU shall return a single negative response with response code \$31 (request out of range) to indicate that freeze frame number \$03 is never supported in the ECU.

Case 7 Request Message: RAMI_ = \$00, FFNR_ = \$FF, RI_ = Not Present in Request

- If neither DTC \$1234 nor DTC \$5678 are stored in non-volatile memory, the ECU shall return a single positive response with FFNR_ = \$FF, RAMI_ = \$00, FFD = Not Present in Response, RI_ = \$0000.
- If either DTC \$1234 or DTC \$5678 are stored in non-volatile memory, the ECU shall respond with a separate positive response message for each freeze frame number for each DTC stored in non-volatile memory.

2.2.2.7.10 inputOutputControlByCommonIdentifier (ref. KWP-GRP-1.5, 9.2.1)

The inputOutputCommonIdentifier values shall refer to Parameter Identifiers (PIDs) specified in the MRDB.

Table 10: Allowable inputOutputControlParameter values

Hex	Description	Man or Opt	Mnemonic
00	ReturnControlToECU This value shall indicate to the server (ECU) that the client (tester) no longer has control about the input signal, internal parameter or output signal referenced by the inputOutputCommonIdentifier.	M	RCTECU
05	FreezeCurrentState This value shall indicate to the server (ECU) that it is requested to freeze the current state of the input signal, internal parameter or output signal referenced by the inputOutputCommonIdentifier.	O	FCS
07	ShortTermAdjustment This value shall indicate to the server (ECU) that it is requested to adjust the input signal, internal parameter or output signal referenced by the inputOutputCommonIdentifier in RAM to the value(s) included in the controlOption parameter(s). (e.g. set Idle Air Control Valve to a specific step number, set pulse width of valve to a specific value/duty cycle).	M	STA

The controlState (CS_) parameter shall only be supported in \$2F request messages with an inputOutputControlParameter of 0x07 "shortTermAdjustment" and is mandatory.

The controlState (CS_) parameter shall only be supported in \$6F positive response messages with an inputOutputControlParameter of 0x05 "freezeCurrentState" or 0x07 "shortTermAdjustment" and is mandatory.

Definitions of all inputOutputCommonIdentifiers, except for those in the supplier reserved range, must be approved by the R&VT EESE Core Network Communication Section. All approved inputOutputCommonIdentifiers and their descriptions are defined and maintained in the MRDB. Each inputOutputCommonIdentifier shall only use the encoding in the MRDB and other encoding values shall not be allowed.

The RRS_ parameter shall consist of a routineStatus byte followed by an optional routineDataResults parameter. The possible values that the routineStatus byte can be are listed in Table 13.

Table 13: Allowable routineStatus Byte values

Hex	Description
00	Routine completed and passed (e.g., no on-demand DTCs detected during self-test)
01	Routine completed and failed (e.g., one or more on-demand DTCs detected during self-test)
02	Routine aborted

The routineDataResults parameter for the local identifiers of \$02 (on-demand self-test) and \$11 (assembly self-test) shall be a list of 2 byte DTCs. The format of the routineDataResults parameter for all execution routines shall be documented in the ECU's SSDS. Table 14 shows the layout of the requestRoutineResultsByLocalIdentifier positive response message.

Table 14: Allowable Routine Status Byte values

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	requestRoutineResultsByLocalIdentifier Pos. Resp. Service Id	M	73	RRRBLIPR
#2	RoutineLocalIdentifier	M	xx	RELI_...
#3	routineStatus	M	00, 01, or 02	} RRS_
:	routineDataResults#1	O	:	
:	:	:	:	
#n	routineDataResults#m	O	xx	

2.2.2.7.14 requestDownload (ref. KWP-GRP-1.5, 11.1.2)

The requestDownload service parameters are listed in Table 15

Table 15: requestDownload Service Message

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	requestDownload Request Service Id	M	34	RD
	transferRequestParameter=[			TRTP_
#2	memoryAddress { MSB of the ECU's address – byte 3 }	M	xx	MA_
#3	memoryAddress { ECU's address – byte 2 }	M	xx	MA_
#4	memoryAddress { ECU's address – byte 1 }	M	xx	MA_
#5	memoryAddress { LSB of the ECU's address – byte 0 }	M	xx	MA_
#6	dataFormatIdentifier	M	xx	DFI
#7	unCompressedMemorySize { High Byte }	M	xx	UCMS
#8	unCompressedMemorySize { Middle Byte }	M	xx	UCMS
#9	unCompressedMemorySize { Low Byte }	M	xx	UCMS

It is recommended that the dataFormatIdentifier (DFI) parameter shall always have the value of \$00, specifying that compression and encryption are not used. However, if compression or encryption is required, the test tool shall not be responsible for any compression or encryption of the data.

2.2.2.7.15 requestDownload Positive Response(ref. KWP-GRP-1.5, 11.1.2)

The requestDownload positive response shall consist of the message as shown in Table 16.

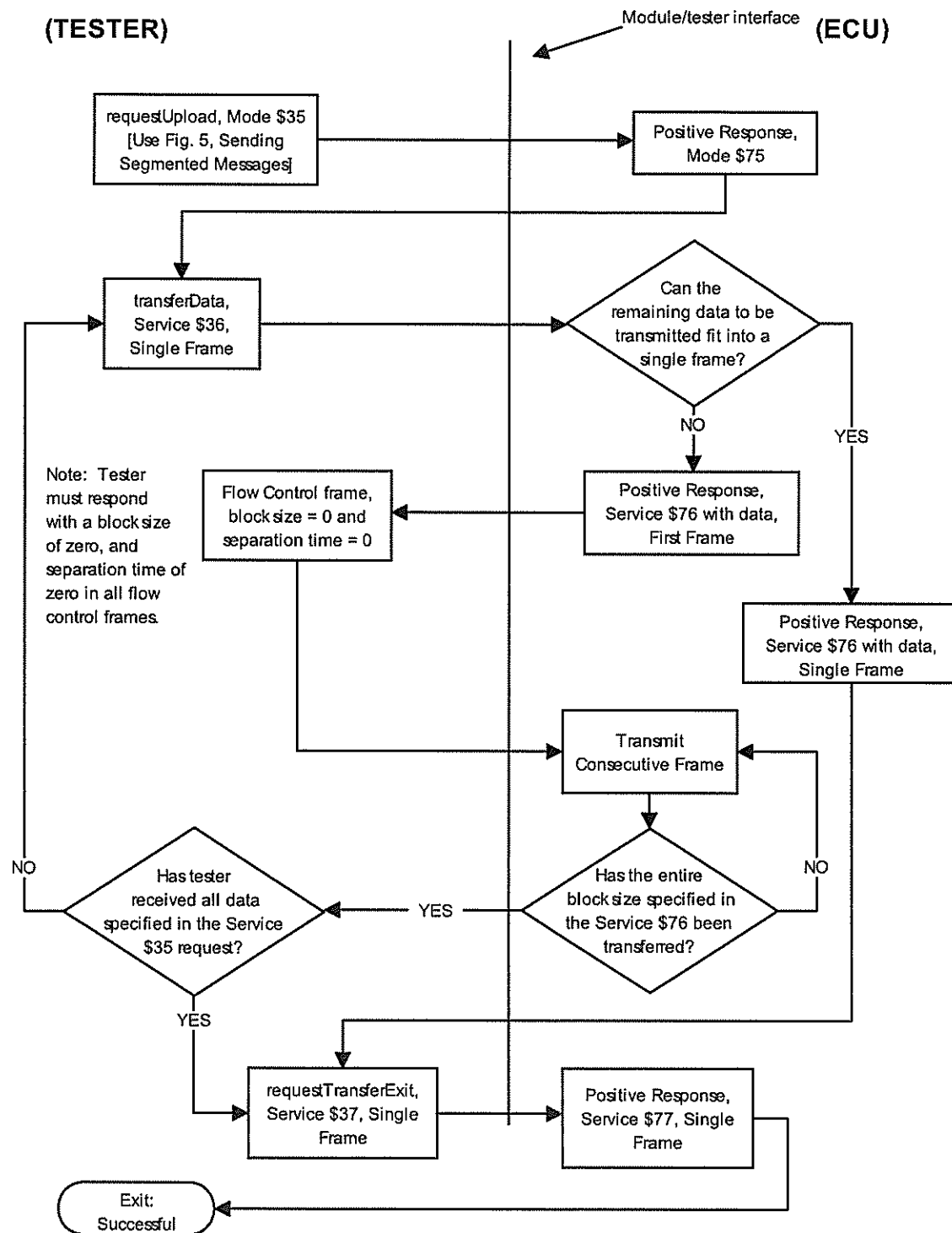


Figure 4: CAN Upload Flowchart

2.2.2.7.19 requestTransferExit (ref. KWP-GRP-1.5, 11.4.2)

The TRTP_ and TREP_ parameters shall not be used for this service. The only parameter in the request and positive response messages for this parameter shall be the RTE parameter.

2.2.2.7.20 testerPresent (ref. KWP-GRP-1.5, 6.4)

The responseRequired parameter shall be supported for all ECUs. Only the values of \$01 (Yes) and \$02 (No) shall be supported for the responseRequired parameter.

Table 19: testerPresent Request Message

Data byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	TesterPresent Request Service Id	M	3E	TP
#2	sub-function = [responseRequired]	M	Xx	RRD_

2.2.2.7.21 diagnosticCommand (ref. P1-v2003.0, 15)

The diagnosticCommand request message is shown in Table 20.

Table 20: diagnosticCommand Request Message

Data Byte	Parameter Name	Cvt	Hex Value
#1	diagnosticCommand Request Service Identifier	M	B1
#2	CommandCommonIdentifier (High Byte)	M	Xx
#3	CommandCommonIdentifier (Low Byte)	M	Xx
#4	Command parameter #1 (Optional – must follow MRDB definition)	O	Xx
#5	Command parameter #2 (Optional – must follow MRDB definition)	O	Xx
#6	Command parameter #3 (Optional – must follow MRDB definition)	O	Xx
#7	Command parameter #4 (Optional – must follow MRDB definition)	O	Xx

Definitions of all CommandCommonIdentifiers, except for those in the supplier reserved range, must be approved by the R&VT EESE Core Network Communication Section. All approved CommandCommonIdentifiers and their descriptions are defined and maintained in the MRDB. Each CommandCommonIdentifier shall only use the length and encoding as specified in the MRDB. Note that the \$B1 service request has a variable length, dependent upon the data length of the requested CommandCommonIdentifier.

The positive response for diagnostic commands is listed in Table 20.

Table 21: diagnosticCommand Positive Response Message

Data Byte	Parameter Name	Cvt	Hex Value
#1	diagnosticCommand Positive Response Service Id	M	F1
#2	commandCommonIdentifier (High Byte)	M	Xx
#3	commandCommonIdentifier (Low Byte)	M	Xx

The negative response for diagnostic commands is listed in Table 22.

is being defined by Memory Address (i.e., DPID Definition Method of 0b10), then RN1 shall correspond to the most significant byte of the memory address, RN2 shall correspond to the 2nd most significant byte, RN3 shall correspond to the 3rd most significant byte, and RN4 shall correspond to the least significant byte of the memory address. If less than four address bytes are supported by an ECU, every Reference Number parameter that is not used shall be set to zero with RN4 always corresponding to the least significant byte of the memory address.

A DPID can be cleared by specifying zero (\$00) as the DPID format byte of a service \$A1 request.

2.2.2.7.22.2 Positive Response Message Definition

Table 25: dynamicallyDefineDiagnosticDataPacket Positive Response Message

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	DynamicallyDefineDiagnosticDataPacket Response Service Id	M	E1	DDDDPPR
#2	Data Packet Identification (DPID) Number	M	xx	DPIDN

2.2.2.7.22.2.1 Positive Response Message Data Parameter Definitions

The DPID Number (DPIDN) parameter is an echo of the DPID Number parameter from the request message.

2.2.2.7.23 requestDiagnosticDataPacket(s) [A0] (ref. P1-v2003.0, 16.10.2)

Table 26, Table 27, and Table 28 list the message parameters for this service.

All references in P1-v2003.0 shall be changed from \$2A to \$A0.

2.2.2.7.23.1 Request Message Definition

Table 26: requestDiagnosticDataPacket(s) Message

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	RequestDiagnosticDataPacket(s) Request Service Id	M	A0	RDDP
#2	Data Rate	M	xx	DR
#3	DPID Number 1	C1	xx	DPID1
#4	DPID Number 2	C2	xx	DPID2
#5	DPID Number 3	C2	xx	DPID3
#6	DPID Number 4	C2	xx	DPID4
#7	DPID Number 5	C2	xx	DPID5

C1 = Parameter is only present if the Data Rate is equal to \$00, \$01, \$02, \$03, \$04 or \$09.

C2 = Parameter may only be present if DPID Number 1 parameter is present. The existence of this parameter depends upon dynamic usage by the requester.

2.2.2.7.23.1.1 Request Message Data Parameter Definitions

The Data Rate contains the action to be performed by this request. The Data Rate shall use only the values and definitions in Table 27.

The positive response to a service \$A0 tester request with a data rate set to \$00, \$05, \$09, or \$0A shall be a single message consisting of only \$E0 for the service byte (no additional data bytes, except padding).

2.2.2.7.23.4 Periodic Positive Responses

Periodic positive diagnostic data packet responses (\$E0) to service \$A0 (request diagnostic data packet) requests shall be transmitted on separate CAN transmission IDs than all other diagnostic responses. These periodic diagnostic IDs are defined and maintained in the MRDB as CAN "Rapid Data" IDs for each ECU where applicable. Some ECUs may require more than one periodic CAN rapid data transmit ID. All diagnostic requests shall still be received on the normal ECU CAN reception ID. Note that non-periodic positive \$E0 responses, such as a positive response to a service \$A0 request with data rate set to \$00, \$01, \$05, \$09, or \$0A shall still be transmitted on the regular CAN transmission ID.

For example, the PCM shall receive service \$A0 requests on CAN ID \$7E0, but shall transmit periodic \$E0 responses on CAN IDs \$6A0 and \$6A1, as defined in the MRDB.

An ECU shall continue to transmit the periodic positive for a particular DPID until one of the following conditions occur:

- 1) The lack of diagnostic dialog timer expires (see section 2.2.3.8)
- 2) A request is received to stop sending the particular DPID (i.e., a service \$A0 request with a data rate of \$00 or \$09 is received for the particular DPID or a service \$A0 request with a data rate of \$05 or \$0A is received).

2.2.2.8 Diagnostic Gateway Requirements

2.2.2.8.1 Purpose / Scope

This section specifies the diagnostic requirements for the design of systems that utilize a gateway module(s). Diagnostic gateway modules translate diagnostic messages from one protocol to another to enable a tester to communicate to modules that only communicate on protocols not supported by a tester; see an example of a gateway topology in Figure 6.

- If the gateway module receives command \$0300 "Gateway State Access" with data byte 5 = \$81, it shall return a Positive Response (service \$F1) as shown in Table 21 and exit the gateway state to the default diagnostic mode.

NOTE: Upon returning to the default diagnostic session, the gateway module shall have 500 milliseconds to reinitialize. During this time period, the gateway module is not required to respond to requests received from a tester.

- If the gateway module receives command \$0300 "Gateway State Access" with data byte 5 not equal to \$00 or \$81, it shall return a Negative Response (service \$7F) of RequestOutOfRange (code \$31).
- If the gateway module receives command \$0300 "Gateway State Access" with data byte 4 containing a value that is not a recognized satellite module address, it shall return a Negative Response (service \$7F) of RequestOutOfRange (code \$31).
- If the gateway module receives a service ID that is not supported in the translation algorithm of the gateway module, it shall return a Negative Response (service \$7F) of ServiceNotSupported (code \$11).

2.2.2.8.3.2 Messages: Gateway to Satellite

Satellite modules that receive translated requests from a gateway module shall handle the various translated requests as follows:

- If the satellite supports the request, it shall formulate a proper response and return it to the gateway module.
- If the satellite receives a service that is not supported, it shall respond with a Negative Response (service \$7F) of ServiceNotSupported (code \$11) to the gateway module.
- If the satellite receives a data byte that is not valid, it shall respond with a Negative Response (service \$7F) of RequestOutOfRange (code \$31) to the gateway module.

2.2.2.8.3.3 Messages: Satellite to Gateway

Gateway modules shall process satellite module responses as follows:

- Correctly formatted satellite responses shall be translated to the communication protocol linking the gateway module and the tester.
- Gateway modules shall ignore a satellite's response if it is an invalid message.

2.2.2.8.4 Gateway State Timing Requirements

The response time between the trailing edge of the last bit of a request message issued by a tester and the leading edge of the response message returned to the tester shall not exceed 275 milliseconds; the tester shall wait 300 milliseconds before logging or indicating an error.

2.2.2.8.5 Satellite Modules

Satellite modules shall adhere to all requirements specified within this specification. Satellite modules are the indirect target of tester request messages when a gateway module is in the gateway state.

NOTE: All tester messages are sent to the gateway module address, not the satellite module address.

2.2.2.8.6 Satellite Module in a Diagnostic Session

All satellite modules shall implement a Lack of Diagnostic Dialog timer (see section 2.2.3.8).

2.2.2.8.7 Satellite Module Mandatory PIDs

All satellite modules shall support all of the (PID/common ID) requirements specified or referenced in this specification and also support (PID/common ID) \$D12D "Node Address."

2.2.2.8.8 Redirecting Communication

2.2.3 Additional Requirements

The requirements in this section are specified for Ford Motor Company and are additional to the requirements referenced in other industry standard documents. These requirements are generic and apply to all ECUs that perform network diagnostics.

2.2.3.1 Diagnostic Services

The following sub-sections contain information on services that are not referenced from the KWP-GRP-1.5 or P1-v2003.0 but are supported for CAN diagnostic communications and services.

2.2.3.1.1 communicationControl (28 hex) service

This service is used to suspend the transmission of normal mode messages in an ECU (i.e., inter-module messages that do not use the diagnostic range of identifiers). Diagnostic messaging, including periodic positive diagnostic data packet responses (rapid data), shall not be affected by this service.

The Lack of Diagnostic Dialog timeout requirements from Section 2.2.3.8 shall apply to this service.

2.2.3.1.1.1 Request Message Definition

Table 30: communicationControl Request Service Message

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	communicationControl Request Service Id	M	28	CC
#2	sub-function = [responseRequired]	M	xx	LEV_RESPREQ
#3	ControlByte	M	xx	CB
#4	communicationType	M	xx	CTP

2.2.3.1.1.1.1 Request Message Sub-Function Parameter LEV_ Definition

The sub-function parameter responseRequired is used by the communicationControl request message to specify if the server shall send a response message; this parameter shall always have the value of \$01 corresponding to the ECU being required to respond to the request.

2.2.3.1.1.1.2 Request Message Data Parameter Definitions

The controlByte parameter contains information on how the ECU shall modify the communication referenced in the communicationType parameter. The information contained in the controlByte parameter is bit oriented and allows for enabling/disabling of the transmission and/or the reception of certain messages of the server. The controlByte (CB) parameter shall use only the values and definitions in Table 31 to control whether to enable or disable communication transmissions in an ECU.

Table 31: controlByte (CB) Parameter Values

Bit number	Bit Definition	'0' Value Definition	'1' Value Definition
0	Control transmission of messages	Enable	Disable
1-7	Reserved by document	N/A	N/A

The communicationType parameter shall indicate which type of messages are either enabled or disabled by this service. Only the values specified in Table 32 shall be supported for the communicationType (CTP) parameter.

Table 32: communicationType (CTP) Parameter Values

Hex Value	Description	Man. or Opt.
01	Application This value references all application-related communication	M
02	NetworkManagement This value references all network management related communication	O

2.2.3.1.3 requestCommonIDScalingMasking (24 hex) service

The requestCommonIDScalingMasking service shall be used to obtain which bits of a bit-mapped common ID are supported.

2.2.3.1.3.1 Service Description

The requestCommonIDScalingMasking request message can be used by a tester to obtain the specific bits of a bit-mapped common ID which are supported for an individual ECU. The ECU being addressed shall respond with a requestCommonIDScalingMasking positive response message or, if unable to perform the request, respond with a requestCommonIDScalingMasking negative response message indicating the reason for the reject.

2.2.3.1.3.2 Request Message Definition

Table 38: requestCommonIDScalingMasking Request Message Definition

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	requestCommonIDScalingMasking Request Service Id	M	24	RCISM
#2	commonIDOffset	M	xx	CIOFF
#3	recordCommonIdentifier (High Byte)	M	xx	RCI_
#4	recordCommonIdentifier (Low Byte)	M	xx	

Only the value of \$FF shall be supported for the commonIDOffset specifying that the offset is not applicable.

2.2.3.1.3.3 Positive Response Message Definition

Table 39: requestCommonIDScalingMasking Positive Response Message

Data Byte	Parameter Name	Cvt	Hex Value	Mnemonic
#1	requestCommonIDScalingMasking Response Service Id	M	64	RCISMPR
#2	commonIDOffset	M	xx	CIOFF
#3	recordCommonIdentifier (High Byte)	M	xx	RCI_
#4	recordCommonIdentifier (Low Byte)	M	xx	
#5	scalingMaskingFormat	M	xx	SMF
#6	scalingMaskingValue#1	M	xx	SMV_
	:	:	xx	:
#n	scalingMaskingValue#m	U	xx	SMV

2.2.3.1.3.3.1 Positive Response Message Data Parameter Definition

Table 40: Response Message Data Parameter Definition

Definition
CommonIDOffset This parameter is an echo of the commonIDOffset parameter from the request message.
RecordCommonIdentifier This parameter is an echo of the recordCommonIdentifier parameter from the request message.

2.2.3.2.3 Bus Off Faults

Bus off failures occur for a variety of reasons as specified in the HS/MS-CAN ECU and Subsystem Requirements Specification and HS/MS-CAN System Level and Distributed Applications Requirements Specification.

DTC \$C073 "Control Module Communication Bus Off" may be set when bus off conditions arise. It is up to the system developer (D&R) to define and implement effective diagnostics associated with DTC \$C073. For example, some ECUs may require that the ECU is bus off for a certain duration before DTC \$C073 is logged. Also, DTC \$C073 should not be logged unless a customer noticeable problem occurs from the bus off condition.

2.2.3.2.4 Private Network Faults

A public network shall be considered any communication bus that is brought out to the J1962 diagnostic connector. A private network shall be considered any communication bus that is not brought out to the J1962 diagnostic connector. An ECU on the public network that is responsible for logging communication DTCs against a private network shall utilize the range of DTCs (\$C028 – \$C072) as allocated in the MRDB. The DTC range consists of faults for five separate private links (Vehicle Communication Bus 'A' through Vehicle Communication Bus 'E'). Each vehicle communication bus DTC group consists of generic, performance, and electrical faults. The type of private link that each vehicle communication bus fault group refers to (e.g., Vehicle Communication Bus 'D') shall be ECU specific. If more than one ECU on the public network logs these communication DTCs against the same physical private network, they shall both use the same vehicle communication group (e.g., both ECUs use Vehicle Communication Bus 'A' faults or both ECUs use Vehicle Communication Bus 'B' faults).

2.2.3.3 Reporting of Logged DTCs

Criteria for logging DTCs in an ECU shall be determined by the module designer unless referenced or specified differently in this specification. All DTCs that are logged in an ECU shall be reported by the readDiagnosticTroubleCodesByStatus service.

2.2.3.4 Negative Response Messages (Service \$7F)

The message data byte structure for negative response messages shall follow the structure specified in Table 41.

Table 41: Negative Response Data Byte Layout

Type	Parameter Name	Hex Value
Data Byte 1	negativeResponse Service Identifier	7F
Data Byte 2	<Service Name> Request Service Identifier	xx
Data Byte 3	responseCode=[KWP2000ResponseCode]	xx=[00-8F, 90-FF]

Refer to Annex A for all applicable negative response codes per diagnostic service.

2.2.3.5 Safety and Damage Prevention Requirements

All diagnostic operations shall have safety and damage protection associated with them. Any operation that can cause a safety hazard or damage the ECU or any of the vehicles system shall have protection associated with it.

For example, performing a service \$2F to drive ABS coils too long can cause damage to the ABS system by burning the coils. For conditions such as this, the ABS module shall only allow the coils to be energized for a finite amount of time to avoid any damage to the ABS sub-system.

of the two following timers expires, then normal ECU control of the operation that was timed out shall resume.

1. **Damage and Safety Timeout:** An ECU needs to suspend the operation of certain timed operations to avoid damage to the ECU or to avoid compromising the safety of any system.
2. **Lack of Diagnostic Dialog:** A 5 second timer shall be supported and reset each time the ECU receives a diagnostic request from a tester. The ECU shall suspend all diagnostic service functionality (see Table 42) upon expiration of the timer. Additionally, if the ECU is not already in the default diagnostic session (\$81) when the timer expires, the ECU shall return to this default session. When receiving a segmented message, the timer shall be suspended when a first frame indication occurs in the ECU and shall be reset whenever a successful reception of a complete message occurs or an error is received from the network layer while receiving the message. When transmitting a segmented message, the timer shall be suspended when the diagnostic message is passed to the network layer for transmission, and shall resume when the network layer indicates the successful transmission of the complete message or indicates any error while transmitting the message. Note that if the Lack of Diagnostic Dialog timer expires when the default diagnostic session (\$81) is active and timed service \$2F (inputOutputControlByCommonID), \$31 (startRoutineByLocalID), or \$B1 (diagnosticCommand) has been successfully performed during this same active diagnostic session, the ECU is allowed to perform a reset to ensure a successful continuation of normal operation.

A **Damage and Safety** timer shall be set up for each service that needs to automatically time out before the **Lack of Diagnostic Dialog** timer expires (e.g., certain diagnosticCommands). The value of each **Damage and Safety** timer that is allocated to any diagnostic operation shall be documented in the ECU's SSDS. The value of the **Damage and Safety** timeout can vary depending on the operation that is initiated by a tester. For example, a specific diagnosticCommand that drives ABS coils may need to be de-energized before the coil burns up.

See Table 42 for the services that shall support the **Lack of Diagnostic Dialog** timeout timer.

Table 42: Services that Support the Lack of Diagnostic Dialog Timer

Hex Value	Diagnostic Services
10	startDiagnosticSession
27	securityAccess
28	communicationControl
2F	InputOutputCtrlByCommonID.
31	startRoutineByLocalIdentifier
85	controlDTCSetting
A0	requestDiagnosticDataPacket
B1	diagnosticCommand

If vehicle speed is available to an ECU, it may suspend all timed diagnostic functionality (see Table 42) and transition to the default diagnostic session when vehicle speed exceeds a value specified in the Subsystem Specific Diagnostic Specification. This shall only be utilized in situations where safety is the primary need for the timeout.

2.2.3.9 Memory Modification and Output Control

WriteMemoryByAddress, writeDataByLocalID and writeDataByCommonID shall be used for writing to memory that shall not be automatically modified by the ECU's software, these services shall not directly modify or control any ECU output signal in any way.

InputOutputControlByCommonIdentifier shall be used for temporarily bypassing inputs (e.g., real world sensors) and for directly controlling output devices only. The substituted value shall be used

2.2.3.15 FNOS Mandatory PIDs

In addition to the referenced mandatory CommonID/PIDs, the FNOS related PIDs as detailed in Table 44 shall be mandatory if the corresponding FNOS component is utilized by the ECU.

Table 44: FNOS CommonIDs/PIDs

CommonID/PID	Description
\$E400	FNOS CAN Driver Version Number
\$E401	FNOS OSEK NM Version Number
\$E402	FNOS NM Junior/Node Management Version Number
\$E403	FNOS Interaction Layer Version Number
\$E404	FNOS Network Initialization Version Number
\$E405	FNOS Transport Layer Version Number
\$E406	FNOS Diagnostics Version Number
\$E407	FNOS Generation Tool Version Number
\$E408	FNOS Bootloader Version Number
\$E409	FNOS Database Version Number

2.2.3.16 Inter-module Communication Gateway

Any ECU functioning as a gateway for application related inter-module communication messages between the public HS-CAN and public MS-CAN network shall support full diagnostics on both networks, but not simultaneously.

2.3.2 Additional Requirements

2.3.2.1 CAN Diagnostic Messages

In all cases, diagnostic frames shall be full-length. The Data Length Code (DLC) parameter that is associated with every CAN frame shall always be 8 for diagnostic CAN frames. Each frame shall be full, consisting of 8 data bytes that can be either transmitted data or pad bytes. Any CAN diagnostic frame with a DLC other than 8 shall be considered invalidly formatted and ignored by the recipient (i.e., no response shall be elicited and no action shall be taken). In the event that the relevant diagnostic information does not completely fill a CAN frame, the remaining bytes within the frame shall be padded with zero (\$00). An ECU shall not reject a received diagnostic request with non-zero pad bytes however.

Note that the term message in CAN refers to the entire set of diagnostic information, whereas frames are the individual units transmitted on CAN bus. For example, if an ECU was to transmit 10 DTCs, the entire set of 10 DTCs with service byte constitutes the message. However, to transmit this data on the CAN bus, the message is broken into frames.

2.3.2.2 CAN Diagnostic Messages Error Handling

2.3.2.2.1 Performance Parameter Definitions (ref ISO 15765-2, 7.6)

Table 46: Segmented Message Performance Parameter Definitions

Parameter	Timeout Value	Performance Requirement Value
N_Bs	75 ms – 120 ms	-
N_Ar + N_Br	-	< 50 ms
N_As + N_Cs	-	< 50 ms
N_Cr	75 ms – 120 ms	-

2.3.2.2.2 Tester to ECU Segmented Message Error Handling

Each segmented message that is initiated by the tester shall have the following error handling associated with it:

Table 47: Tester to ECU Segmented Error Handling

Error Condition	Error Handling
N_Ar + N_Br parameter exceeded	Product development shall fail the ECU during testing and fix the problem.
N_Bs timeout exceeded	Test equipment shall assume the ECU will not respond to the request. The tester can initiate other requests.
N_Cr timeout exceeded	ECU shall send a negative response to the first frame with response code of \$72 "transferAborted".

2.3.2.2.3 ECU to tester Segmented Message Error Handling

Each segmented message that is initiated by the ECU shall have the following error handling associated with it:

Table 48: ECU to Tester Segmented Error Handling

Error Condition	Error Handling
N_Bs timeout exceeded	The ECU shall timeout and terminate the remainder of the message.
N_As + N_Cs parameter exceeded	Product development shall fail the ECU during testing and fix the problem.
N_Cr timeout exceeded	Test equipment shall assume the ECU will not send any more frames within this message. The tester can initiate other requests.

Annex A

A.1 Service Response Code Tables

This annex contains tables of every service available for use in this document. Each table lists all of the allowable response codes associated with each service. No additional response codes shall be applied to any service if they are not shown in each service's table except for the general ones that apply to all services.

A.2 General Response Codes

General response codes may be used for any service where necessary. As an example, the general response code of busy (\$21) shall be supported by an ECU in response to a request to obtain routine results while the execution routine is running, but may never be used by an ECU in response to a CommonID (PID) request.

The general response code of busy-repeatRequest (\$21) is returned by the ECU if the ECU is temporarily too busy to perform the requested operation. The response code indicates that the requested action was not performed, but that a later repetition of the request will eventually result in a positive response. In this circumstance the tester may perform repetition of the "identical request message" or "another request message". The length of time an ECU may respond with a negative response code of \$21 to a tester request shall always be finite.

The general response code of reqCorrectlyRevld-RspPending (\$78) is returned by the ECU if the time required to complete the request takes longer than 50ms. See Section 2.2.2.6 for more information related to response code \$78. General response code \$78 shall only be used when necessary and only under the following conditions:

- Long duration memory writes when responding to diagnostic service \$14, \$2E, \$3D, \$3B, or \$36
- Checksum calculations and/or application validation when responding to diagnostic service \$11 or \$37
- Erasure of flash memory when responding to diagnostic service \$B1 with a CommandCommonID of \$00B2 (Flash Memory Erase)

In all other cases other than described above, an ECU is not allowed to respond with a negative response code of \$78 unless explicit approval is obtained by EESE Core Network Communications.

Response code \$72 handling shall be performed as specified in Section 2.3.2.2.2 for segmented messages initiated by the test equipment. In addition, response code \$72 shall be used when an ECU receives a consecutive frame from a tester with a sequence number not in the correct order.

If an ECU receives a first frame from a tester with a first frame data length (FF_DL) less than eight, the ECU shall respond with a negative response code of illegalByteCountInBlockTransfer (\$75).

A.2.4 (\$14) clearDiagnosticInformation

Hex	Description
12	SubFunctionNotSupported-InvalidFormat 1. The length of the message is wrong 2. All required data byte(s) were not supplied 3. Extra data byte(s) were supplied
22	ConditionsNotCorrect 1. Internal conditions in the ECU prevent clearing of DTCs 2. Not in the correct diagnostic session
31	RequestOutOfRange The requested groupOfDiagnosticInformation parameter is not supported by the ECU

A.2.5 (\$18) readDTCByStatus

Hex	Description
12	SubFunctionNotSupported-InvalidFormat 1. The length of the message is wrong 2. All required data byte(s) were not supplied 3. Extra data byte(s) were supplied 4. The requested statusOfDTC parameter is not supported by the ECU
22	ConditionsNotCorrect Not in the correct diagnostic session
31	RequestOutOfRange The requested groupOfDTC parameter is not supported by the ECU

A.2.6 (\$21) readDataByLocalID

Hex	Description
11	ServiceNotSupported This service is not supported by the ECU under any circumstances
12	SubFunctionNotSupported-InvalidFormat 1. The length of the message is wrong 2. All required data byte(s) were not supplied 3. Extra data byte(s) were supplied
22	ConditionsNotCorrect Not in the correct diagnostic session
31	RequestOutOfRange The requested local ID is not supported
33	SecurityAccessDenied-securityAccessRequested The module has not been unlocked and needs to be in order for this request to work

A.2.10 (\$27) securityAccess

Hex	Description
11	ServiceNotSupported This service is not supported by the ECU under any circumstances
12	SubFunctionNotSupported-InvalidFormat - The length of the message is wrong - All required data byte(s) were not supplied - Extra data byte(s) were supplied
22	ConditionsNotCorrect - Not in the correct diagnostic session - A 'sendkey' parameter is received by the ECU without a valid or active seed present for that security level.
31	RequestOutOfRange The security level requested is not supported by the ECU
35	InvalidKey The key is incorrect

A.2.11 (\$28) communicationControl

Hex	Description
12	SubFunctionNotSupported-InvalidFormat - The length of the message is wrong - All required data byte(s) were not supplied - Extra data byte(s) were supplied - The requested responseRequired parameter is not supported by the ECU
22	ConditionsNotCorrect - Not in the correct diagnostic session - Some vehicle conditions exist that prohibit this service from being executed
31	RequestOutOfRange - The controlByte parameter is out of range - The communicationType parameter is out of range
33	SecurityAccessDenied-securityAccessRequested The module has not been unlocked and needs to be in order for this request to work

A.2.12 (\$2E) writeDataByCommonId

Hex	Description
11	ServiceNotSupported This service is not supported by the ECU under any circumstances
12	SubFunctionNotSupported-InvalidFormat - The length of the message is wrong - All required data byte(s) were not supplied - Extra data byte(s) were supplied
22	ConditionsNotCorrect Not in the correct diagnostic session
31	RequestOutOfRange - The requested common ID is not supported - Any data transmitted in the request message after the common ID is invalid
33	SecurityAccessDenied-securityAccessRequested The module has not been unlocked and needs to be in order for this request to work

A.2.16 (\$33) requestRoutineResultsByLocalID

Hex	Description
12	SubFunctionNotSupported-InvalidFormat 1. The length of the message is wrong 2. All required data byte(s) were not supplied 3. Extra data byte(s) were supplied
22	ConditionsNotCorrect 1. The execution routine is supported but has not been ran in the current diagnostic session 2. Not in the correct diagnostic session
31	RequestOutOfRange This local ID is not supported
33	SecurityAccessDenied-securityAccessRequested The module has not been unlocked and needs to be in order for this request to work

A.2.17 (\$34) requestDownload

Hex	Description
10	GeneralReject Some internal condition is preventing a download from occurring (e.g., flash memory is not blank)
11	ServiceNotSupported This service is not supported by the ECU under any circumstances
12	SubFunctionNotSupported-InvalidFormat 1. The length of the message is wrong 2. All required data byte(s) were not supplied 3. Extra data byte(s) were supplied
22	ConditionsNotCorrect Not in the correct diagnostic session
31	RequestOutOfRange The address, length, or dataFormatIdentifier is not correct
33	SecurityAccessDenied-securityAccessRequested The module has not been unlocked and needs to be in order for this request to work

A.2.18 (\$35) requestUpload

Hex	Description
11	ServiceNotSupported This service is not supported by the ECU under any circumstances
12	SubFunctionNotSupported-InvalidFormat 1. The length of the message is wrong 2. All required data byte(s) were not supplied 3. Extra data byte(s) were supplied
22	ConditionsNotCorrect Not in the correct diagnostic session
31	RequestOutOfRange The address, length, or dataFormatIdentifier is not correct
33	SecurityAccessDenied-securityAccessRequested The module has not been unlocked and needs to be in order for this request to work

A.2.21 (\$3B) writeDataByLocalId

Hex	Description
10	GeneralReject Some internal error occurred (e.g., battery voltage too low)
11	ServiceNotSupported This service is not supported by the ECU under any circumstances
12	SubFunctionNotSupported-InvalidFormat 1. The length of the message is wrong 2. All required data byte(s) were not supplied 3. Extra data byte(s) were supplied
22	ConditionsNotCorrect Not in the correct diagnostic session
31	RequestOutOfRange 1. This local ID is not supported 2. Any data transmitted in the request message after the local ID is invalid
33	SecurityAccessDenied-securityAccessRequested The module has not been unlocked and needs to be in order for this request to work

A.2.22 (\$3D) writeMemoryByAddress

Hex	Description
11	ServiceNotSupported This service is not supported by the ECU under any circumstances
12	SubFunctionNotSupported-InvalidFormat 1. The length of the message is wrong 2. All required data byte(s) were not supplied 3. Extra data byte(s) were supplied
22	ConditionsNotCorrect Not in the correct diagnostic session
31	RequestOutOfRange This address or length to write to is not supported
33	SecurityAccessDenied-securityAccessRequested The module has not been unlocked and needs to be in order for this request to work

A.2.23 (\$3E) testerPresent

Hex	Description
12	SubFunctionNotSupported-InvalidFormat 1. The length of the message is wrong 2. Extra data byte(s) were supplied 3. The requested responseRequired parameter is not supported by the ECU

A.2.27 (\$B1) diagnosticCommand

Hex	Description
10	GeneralReject Some internal error occurred
11	ServiceNotSupported This service is not supported by the ECU under any circumstances
12	SubFunctionNotSupported-InvalidFormat 1. The length of the message is wrong 2. All required data byte(s) were not supplied 3. Extra data byte(s) were supplied
22	ConditionsNotCorrect 1. Not in the correct diagnostic session 2. The necessary code to complete this command functionality has not been downloaded yet (e.g., the secondary bootloader is not present when a request is received to erase flash memory). 3. All prerequisites for the execution of the diagnosticCommand are not satisfied
31	RequestOutOfRange 1. The requested Command Common Identifier is not supported by the ECU 2. The requested Command Parameter is not supported by the ECU
33	SecurityAccessDenied-securityAccessRequested The module has not been unlocked and needs to be in order for this request to work